

# Léonel Vodounou

Mathematical Engineer · Aspiring AI Researcher

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## Research Interests

Large language models and the limits of autoregressive reasoning; neuro-symbolic learning (Logic Tensor Networks); mathematical foundations of deep learning: probability, stochastic modeling and optimization. I am interested in understanding models down to their equations and rebuilding them from first principles.

## Education

**Engineering Degree in Mathematical Engineering and Modeling** 2022 – 2025  
*National School of Mathematical Engineering and Modeling (ENSGMM), UNSTIM* Benin

- Specialization in random modeling, statistics and machine learning.

**Preparatory Classes for Engineering Studies (MPSI/PCSI)** 2020 – 2022  
*National Higher Institute of Preparatory Classes (INSPEI), UNSTIM* Benin

- Intensive training in mathematics, physics and engineering sciences.

## Research Projects

**Logic Tensor Networks: presentation and case study** 2026  
[github.com/vleonel-junior/logic-tensor-networks](https://github.com/vleonel-junior/logic-tensor-networks)

- Presentation of the neuro-symbolic LTN framework: grounding of logical symbols as tensors, differentiable connectives and quantifiers, knowledge-base satisfaction as a training objective.
- Case study with LTNtorch: semi-supervised MNIST digit recognition learned only from the sum of two digits, compared with a purely supervised baseline.

**Building a Large Language Model from Scratch: reading notes** 2026  
*Bilingual (EN/FR) chapter-by-chapter notes on S. Raschka's book, [published online](#)*

- From-scratch PyTorch implementations of tokenization, byte pair encoding, data sampling, embeddings, self-attention, causal and multi-head attention.
- Complementary derivations: softmax Jacobian, the  $1/\sqrt{d_k}$  scaling factor, equivalence of causal masking strategies, variance under weight initialization.

**Earlier projects** 2025  
*Mathematical programming and quantitative finance*

- **Gomory cutting-plane method**: pure Python solver for integer linear programs with exact fractional arithmetic.
- **Black-Scholes**: analytical solution through the heat equation, pricing formula and Delta of a European call.

## Writing

- *Why LLMs Struggle to Reason: The Limits of Autoregression*, blog article, July 2026.
- *Descriptive Statistics*, interactive blog article, March 2026.
- All articles in English and French at [vleonel-junior.github.io/blog](https://vleonel-junior.github.io/blog).

## Experience

**Data Scientist Intern (end-of-studies internship)** Sep 2025 – Dec 2025  
*LKA Services* Benin

- Designed data processing and analysis solutions, automated classification processes and improved data management to support decision-making.

**Data Scientist Intern** Aug 2024 – Oct 2024  
*LESCAL, Laboratory for Statistical Studies and Application & Software Design* Benin

- Contributed to statistical studies and the design of data-driven applications.

## Conferences and Communities

**Deep Learning Indaba 2026** Aug 2026  
*Participant* Lagos, Nigeria

**Deep Learning IndabaX Benin 2026** Sep 2026  
*In-person participant* Benin

Teaching and Outreach

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Volunteer trainer: AI for artisans and entrepreneurs</b><br><i>iSHEERO, Sèmè City Open Park</i>                                                                                             | Jul 2026<br>Benin |
| <ul style="list-style-type: none"><li>Co-led a hands-on workshop on generative AI tools for about twenty artisans and local entrepreneurs, built around their day-to-day challenges.</li></ul> |                   |

Training and Certifications

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|-----------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Machine Learning Engineer program</b><br><i>iSheero × DataCamp, 12-month program with applied projects</i>                     | 2026     |
| <b>Data Scientist</b><br><i>Africa Tech Up Tour / OpenClassrooms</i>                                                              | 2024     |
| <b>Introduction to High-Performance Computing</b><br><i>Supercomputer access, job management with SLURM, parallel programming</i> | Jun 2024 |

Skills

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|-------------------------|------------------------------------------------------------------------------------------|
| <b>Mathematics</b>      | Probability and statistics, stochastic modeling, optimization, numerical analysis (PDEs) |
| <b>Machine learning</b> | PyTorch, transformers and attention mechanisms, LTNtorch                                 |
| <b>Programming</b>      | Python, R, MATLAB, L <sup>A</sup> T <sub>E</sub> X, Git                                  |
| <b>Languages</b>        | French (native), Fon (native), English (fluent)                                          |